CHAPTER 14

SEMICONDUCTOR ELECTRONICS: MATERIALS, DEVICES AND SIMPLE CIRCUITS

CHAPTER ESSENCE AT A GLANCE:
Energy band theory, p-n junction diode forward and reverse characteristics, half and full-wave rectifiers, and Zener voltage regulation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Energy Band Classification:** Solids are classified by forbidden energy gaps between valence and conduction bands: large gaps in insulators, overlapping bands in conductors, and small gaps in semiconductors.

• **Intrinsic vs Extrinsic Semiconductors:** Pure semiconductors conduct through thermally generated equal electron-hole pairs; pentavalent doping produces n-type, while trivalent doping produces p-type semiconductors.

• **p-n Junction Depletion Barrier:** Diffusion of charge carriers across the interface creates a stationary space-charge depletion region with a built-in potential barrier.

• **Semiconductor Diode Rectification:** A p-n junction diode conducts heavily under forward bias and blocks current under reverse bias, converting alternating current into pulsating direct current.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$n_i^2 = n_e \cdot n_h$$

Mass Action Law for semiconductors in thermal equilibrium

$$\alpha = \frac{\beta}{1+\beta} \quad \text{and} \quad \beta = \frac{\alpha}{1-\alpha}$$

Common emitter and common base current gain relations

$$Y = \overline{A \cdot B} \quad (\text{NAND}), \quad Y = \overline{A + B} \quad (\text{NOR})$$

Boolean expressions for universal logic gates

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Evaluators deduct 1 Mark if input AC waveform and output pulsating DC waveform are not drawn together!

• **Alert:** [Valuation Rubric] Draw center-tapped transformer, two diodes D_1 and D_2, and load resistor R L .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Energy bands in solids: valence band, conduction band, and forbidden energy gap E g [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Energy Band Concepts]** 1. Valence Band (VB): Energy band formed by valence electrons of atoms. Completely or partially filled at 0 K, never empty. Does not contribute to conduction unless electrons gain energy. [1 Mark]

• 2. Conduction Band (CB): Energy band formed by free conduction electrons. Lowest unfilled band. At 0 K, it is completely empty in insulators and semiconductors. [0.5 Mark]

• 3. Forbidden Energy Gap (Bandgap, E g): The energy separation between the top of valence band (E v) and bottom of conduction band (E c). No electron quantum states can exist in this gap. [0.5 Mark]

• **Key Exam Result:** E g = E c - E v; Determines electrical conductivity of solid

2. Classification of metals, insulators, and semiconductors on the basis of energy band gap [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Classification on Energy Band Gap E g]** 1. Metals or Conductors: Conduction band and valence band either overlap, or valence band is partially filled (E g approx. 0 eV). An abundance of free electrons is available for conduction even at 0 K (Resistivity 10^{-2} - 10^{-8} Ω m). [1 Mark]

• 2. Insulators: Large forbidden energy gap (E g > 3 eV; for Diamond E g approx. 5.4 eV). Thermal excitation at room temperature is insufficient to jump this wide gap. Conduction band remains completely empty (Resistivity 10^{11} - 10^{19} Ω m). [1 Mark]

• 3. Semiconductors: Finite small energy gap (E g < 3 eV; for Silicon E g = 1.1 eV, Germanium E g = 0.7 eV). At 0 K, behaves as insulator; at room temperature, some valence electrons bridge gap, enabling moderate conduction. [1 Mark]

• **Key Exam Result:** Metals (E g approx. 0) | Semiconductors (0 < E g < 3 eV) | Insulators (E g > 3 eV)

3. Intrinsic semiconductors: silicon and germanium, electron-hole pairs, electrical conductivity [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Intrinsic Semiconductors]** Pure crystalline semiconductors (Silicon, Germanium) free from chemical impurities. Each Si atom has 4 valence electrons forming 4 covalent bonds with neighboring atoms. [1 Mark]

• **[Conduction Mechanism & Electron-Hole Pairs]** At 0 K, all covalent bonds are intact (acts as perfect insulator). At room temperature (T > 0 K), thermal energy breaks a few covalent bonds, creating free electrons in conduction band and leaving behind vacancies in valence band called HOLES (effective positive charge +e). Electron density equals hole density: n e = n h = n i (intrinsic carrier concentration). Total electrical current is sum of electron and hole currents: I = I e + I h = e A (n e v e + n h v h). [1 Mark]

• **Key Exam Result:** Intrinsic: n e = n h = n i; Total Current I = I e + I h

4. Extrinsic semiconductors: n-type (pentavalent doping) and p-type (trivalent doping) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Extrinsic Semiconductors & Doping Types]** Extrinsic semiconductor is formed by deliberately adding a tiny controlled amount of specific impurity atoms (dopant) to pure semiconductor to dramatically boost conductivity. [1 Mark]

• 1. n-type: Doped with Pentavalent impurities (P, As, Sb - Donor atoms). Four electrons form covalent bonds; fifth electron is weakly bound (energy ~0.05 eV below CB) and ionized at room temperature. Majority carriers: ELECTRONS (n e >> n h). [1 Mark]

• 2. p-type: Doped with Trivalent impurities (B, Al, In - Acceptor atoms). Three valence electrons form bonds, leaving one vacancy (hole) that accepts neighbor electrons. Majority carriers: HOLES (n h >> n e). Both types remain electrically neutral: n e n h = n_i^2 (Mass Action Law). [1 Mark]

• **Key Exam Result:** n-type (Pentavalent dopant, electrons majority) | p-type (Trivalent dopant, holes majority)

5. p-n junction formation: diffusion current, drift current, depletion layer, and barrier potential [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[p-n Junction Formation & Processes]** When p-type and n-type semiconductor regions are joined in a single crystal: 1. Diffusion Current: High hole concentration in p-region diffuses into n-region; high electron concentration in n-region diffuses into p-region. [1 Mark]

• 2. Depletion Layer & Barrier Potential: Recombination of diffusing electrons and holes near boundary leaves unneutralized immobile ionized donor ions (+ve) on n-side and acceptor ions (-ve) on p-side. This narrow region devoid of free mobile charge carriers is the Depletion Layer (~0.5 μm). The internal electric field directed from n to p creates a Barrier Potential V b (~0.7 V for Si, 0.3 V for Ge) that halts further majority carrier diffusion. 3. Drift Current: Minorities pushed across junction by internal field; in equilibrium: I diff = I drift implies net current = 0. [2 Marks]

• **Key Exam Result:** Depletion layer contains immobile ions; Barrier V b = 0.7 V (Si), 0.3 V (Ge)

6. Numerical on dynamic resistance of p-n junction diode from V-I curve \ΔV divided by \ΔI [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula for Dynamic Resistance]** Dynamic resistance r d = ΔV divided by ΔI. [1 Mark]

• **[Step 2: Substitution & Evaluation]** In forward bias: voltage increases from 0.70 V to 0.75 V (ΔV = 0.05 V), and current increases from 10 mA to 20 mA (ΔI = 10 mA = 10×10^{-3} A). Forward dynamic resistance: r f = ΔV divided by ΔI = 0.05 V divided by (10×10^{-3} A) = 5.0 Ω. In reverse bias: voltage changes from 10 V to 20 V (ΔV = 10 V), and reverse current changes from 1.0 μA to 1.2 μA (ΔI = 0.2 μA = 0.2×10^{-6} A). Reverse dynamic resistance: r r = 10 V divided by (0.2×10^{-6} A) = 5.0×10^{7} Ω = 50 MΩ. [1 Mark]

• **Key Exam Result:** Forward resistance r f = 5.0 Ω; Reverse resistance r r = 50 MΩ (ratio 1 : 10^{7})

7. Numerical on output frequency of half-wave and full-wave rectifier for 50 Hz AC input [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formulas]** Input AC frequency f in = 50 Hz (period T = 1 divided by 50 = 0.02 s). 1. Half-Wave Rectifier: Conduction occurs only during one half-cycle per period; output pulses repeat every 0.02 s. Output frequency: f out = f in = 50 Hz. [1 Mark]

• 2. Full-Wave Rectifier: Conduction occurs during both positive and negative half-cycles; output pulses repeat twice per period (every 0.01 s). Output frequency: f out = 2 × f in = 2 × 50 Hz = 100 Hz. [1 Mark]

• **Key Exam Result:** Half-Wave output frequency = 50 Hz | Full-Wave output frequency = 100 Hz